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UNO R4 WiFi Network Examples

Discover examples compatible with the WiFi library included in the UNO R4 Board Package.

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The [Arduino UNO R4 WiFi](#) has a built in ESP32-S3 module that enables you to connect to Wi-Fi® networks, and perform network operations. Protocols including HTTPS, MQTT, UDP are tested and supported, and in this article, you will find a number of examples that will get you started.

Wi-Fi® support is enabled via the built-in `WiFiS3` library that is shipped with the [Arduino UNO R4 Board Package](#). Installing the Board Package automatically installs the `WiFiS3` library.

The easiest way to connect your board to the Internet is via the [Arduino Cloud](#) platform.

i Here you can configure, program, monitor and synchronize your devices without having to write any networking code.

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Examples

Examples listed in this section have been tested and verified to work. Most examples require you to input the `SSID` and `PASSWORD` for your local Wi-Fi® network. As a standard practice, in all of our examples, we store this in a separate header file (called `arduino_secrets.h`).

You will need to create this file, or remove the

```
#include  
"arduino_secrets.h"
```

file at the top of each example.

The file should contain:

```
1 //arduino_secrets.h  
2 #define SECRET_SSID "  
3 #define SECRET_PASS "
```

Storing your network & password in a separate



file minimizes the risk of you accidentally sharing your Wi-Fi® credentials.

Access Point



Code Source on
[Github](#)

```
1  /*
2   WiFi Web Server
3
4   A simple web server
5   This sketch will
6   It will then launch
7   to the Serial Monitor
8   to turn on and on
9
10  If the IP address is
11     http://yourAddress
12     http://yourAddress
13
14  created 25 Nov 2013
15  by Tom Igoe
16  adapted to WiFi
17
18  Find the full UNO
19  https://docs.arduino.cc
20  */
21
22  #include "WiFi.h"
23
24  #include "arduino.h"
25
26  ///////////please enter
27  char ssid[] = "SECRET";
28  char pass[] = "SECRET";
```

Connect With WPA



Code Source on
[Github](#)

```
1  /*
2   This example con
3   Then it prints th
4   the IP address of
5
6   created 13 July :
7   by dlf (Metodo2 :
8   modified 31 May :
9   by Tom Igoe
10
11   Find the full UN
12   https://docs.ard
13   */
14   #include <WiFiS3.h>
15
16   #include "arduino_
17   //please ente
18   char ssid[] = SECR
19   char pass[] = SECR
20   int status = WL_ID
21
22   void setup() {
23     //Initialize ser
24     Serial.begin(9600
25     while (!Serial)
26       ; // wait for :
27   }
28
29   --
```

Scan Networks



Code Source on
[Github](#)

```
1  /*
2   This example pr
3   scans for availa
4   Every ten second
5   connect to any n
6
7   Circuit:
8   * Uno R4 Wi-Fi
9
10  created 13 July :
11  by dlf (Metodo2 :
12  modified 21 Junn
13  by Tom Igoe and .
14
15  Find the full UNO
16  https://docs.ard
17  */
18
19
20
21  #include <WiFiS3.h>
22
23  void setup() {
24    //Initialize ser
25    Serial.begin(9600
26    while (!Serial)
27      ; // wait for :
28  }
29
30  --
```

Scan Networks (Advanced)



Code Source on
[Github](#)

```
1  /*
2   This example pr
3   scans for availa
4   Every ten second
5   connect to any n
6   BSSID and WiFi cl
7
8   Circuit:
9   * Uno R4 WiFi
10
11  This example is l
12
13  created 1 Mar 20
14  by Arturo Guadal
15
16  Find the full UNO
17  https://docs.ard
18  */
19
20
21
22  #include <WiFiS3.h>
23
24  void setup() {
25    //Initialize ser
26    Serial.begin(9600
27    while (!Serial)
28      ; // wait for :
29  }
```

Simple Webserver



Code Source on
[Github](#)

```
1  /*
2   WiFi Web Server
3
4   A simple web server
5   This sketch will
6   to the Serial Monitor
7   to turn on and off
8
9   If the IP address
10  http://yourAddress
11  http://yourAddress
12
13  This example is for
14  WEP or WPA, change
15
16  Circuit:
17  * Board with NINA
18  * LED attached to
19
20  created 25 Nov 2015
21  by Tom Igoe
22
23  Find the full UNO
24  https://docs.arduino.cc
25  */
26
27  #include "WiFiS3.h"
28
```

Wi-Fi® Chat Server



Code Source on
[Github](#)

```
1  /*
2   Chat  Server
3
4   A simple server
5   connected client:
6   You can see the
7
8   This example is v
9   WEP or WPA, chang
10
11
12  Circuit:
13  * Board with NIN
14
15  created 18 Dec 20
16  by David A. Mell
17  modified 31 May
18  by Tom Igoe
19
20  Find the full UN
21  https://docs.ard
22  */
23
24
25  #include <WiFiS3.h>
26
27  #include "arduino_
28  //please ente
```

Wi-Fi® UDP NTP Client



Code Source on
[Github](#)


```
1  /*
2   Udp NTP Client
3
4   Get the time from
5   Demonstrates use
6   For more on NTP
7   see http://en.wil
8
9   created 4 Sep 20
10  by Michael Margo
11  modified 9 Apr 20
12  by Tom Igoe
13  modified May, 4tl
14  by Daniele Aimo
15
16  This code is in
17
18  Find the full UNO
19  https://docs.ard
20  */
21
22
23  #include <WiFiS3.h>
24
25  int status = WL_ID
26  #include "arduino_
27  //please ente
28  char ssid[] = SECR
--
```

Wi-Fi® UDP Send Receive String



Code Source on
[Github](#)

```
1  /*
2   WiFi UDP Send and
3
4   This sketch waits
5   When a packet is
6
7   created 30 Deceml
8   by dlf (Metodo2
9
10  Find the full UN
11  https://docs.ard
12  */
13
14
15  #include <WiFiS3.h>
16
17  int status = WL_ID
18  #include "arduino_
19  //please ente
20  char ssid[] = SECR
21  char pass[] = SECR
22  int keyIndex = 0;
23
24  unsigned int local
25
26  char packetBuffer[
27  char ReplyBuffer[
28
29  ...
```

Wi-Fi® Web Client



Code Source on
[Github](#)

```
1  /*
2   Web client
3
4   This sketch connects to a web server
5   using the WiFi module.
6
7   This example is for the Arduino Uno
8   WEP or WPA, change the key and SSID
9
10  This example is for the Arduino Uno
11  WEP or WPA, change the key and SSID
12
13
14  created 13 July 2015
15  by dlf (Metodo24)
16  modified 31 May 2016
17  by Tom Igoe
18
19  Find the full UNO WiFi module pinout
20  https://docs.arduino.cc/hardware/wifi
21  */
22
23
24  #include "WiFiS3.h"
25
26
27  #include "arduino.h"
28
29  // ...
```

Wi-Fi® Web Client Repeating



Code Source on
[Github](#)

```
1  /*
2   Repeating WiFi W
3
4   This sketch connec
5   using a WiFi equip
6
7   created 23 April :
8   modified 31 May 20
9   by Tom Igoe
10  modified 13 Jan 20
11  by Federico Vanza
12
13  This code is in tl
14
15   Find the full UNO
16   https://docs.ard
17  */
18
19  #include "WiFiS3.h
20
21
22  #include "arduino_
23  //please ente
24  char ssid[] = SECR
25  char pass[] = SECR
26  int keyIndex = 0;
27
28  int status = WL_IDI
--
```

Wi-Fi® Web Client SSL



Code Source on
[Github](#)

```
1  /*
2   TLS WiFi Web cli
3
4   Board CA Root ce
5   https://github.co
6
7   Find the full UNO
8   https://docs.ard
9  */
10
11  #include "WiFiS3.h
12  #include "WiFiSSLC
13  #include "IPAdres
14
15  #include "arduino_
16
17  //////////please ente
18  char ssid[] = SECR
19  char pass[] = SECR
20
21  int status = WL_ID
22  // if you don't wai
23  // use the numeric
24  //IPAddress server
25  char server[] = "w
26
27  // Initialize the l
28  // with the IP add
--
```

Wi-Fi® Web Server



Code Source on
[Github](#)

```
1  /*
2   WiFi Web Server
3
4   A simple web server
5
6   This example is w
7   WEP or WPA, chang
8
9   Circuit:
10  * Analog inputs a
11
12  created 13 July 20
13  by dlf (Metodo2 s
14  modified 31 May 20
15  by Tom Igoe
16
17
18  Find the full UNO
19  https://docs.ardi
20  */
21
22  #include "WiFiS3.h
23
24
25
26  #include "arduino_
27  //////////please ente
28  char ssid[] = SECR
```

Suggest changes

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